Contents

1 General **2**
1.1 Syntax, Aufbau . . . 2
1.1.1 Dokumentklassen . . . 2
1.1.2 Packages . . . 2
1.1.3 Makros . . . 2
1.1.4 Environments . . . 2
1.1.5 Counter . . . 2
1.1.6 Hyperref . . . 2
1.2 Geometrie und Größen . . . 2
1.2.1 More geometries . . . 3
1.2.2 Farben, Highlight . . . 3
1.2.3 Einzug . . . 3
1.2.4 Textlänge kürzen . . . 3
1.3 Datum . . . 3
1.4 Kopf- und Fußzeile . . . 3
1.4.1 Seitenzahlanzeige . . . 4
1.5 Überschriften . . . 4

2 Images **5**

3 Math **6**
3.1 Special characters . . . 6
3.2 Math-Operator . . . 6
3.3 Equations . . . 6
3.3.1 Multiline Equations . . . 6
3.3.2 Aligned Equations . . . 6
3.4 Theorems . . . 7
3.5 Matrizen . . . 7
3.6 Math alignment . . . 7

4 Auflistungen **9**
4.1 `enumitem` . . . 9
4.1.1 Einzug in Liste . . . 9
4.1.2 Andere Symbole, Abstände, Horizontale Bulletpoints . . . 9
4.1.3 Liste mit anderer Zahl anfangen . . . 9
4.2 Tabelle . . . 10

5 Bibliography **11**

6 Sourcecode **12**

7 Fun Extras **12**
7.1 If-Then Unterscheidungen . . . 12
7.2 Inhaltsverzeichnis / Table of contents . . . 13
7.2.1 Veringern des vertikalen Abstandes im / vorm Inhaltsverzeichnis . . . 13
7.2.2 Eigene Verzeichnisse . . . 13

Kapitel 1

General

TODO: "Highlight in listing-mode"

1.1 Syntax, Aufbau

1.1.1 Dokumentklassen

1.1.2 Packages

Custom / modified packages (.sty) can be imported using the `\usepackage{package}` command when the file.sty is in the same directory as the main .tex file.

1.1.3 Makros

1.1.4 Environments

```
\usepackage{amsthm,amsmath,amssymb}

\newcommand*{\claimname}{claim.}
\newenvironment{claim}[1][\claimname]{\begin{proof}[#1]\renewcommand*{\qedsymbol}{\(\boxtimes\)}}{\end{proof
```

1.1.5 Counter

1.1.6 Hyperref

`\usepackage{hyperref}`

1.2 Geometrie und Größen

```
\usepackage{geometry}

\geometry{
    a4paper,
    left=20mm,
    right=20mm,
    bottom=20mm,
    top=22mm
}
% pt...point
% pc...pica (=12pt)
% bp...bigpoint(=72pt,ca 1in)
% mm
% cm
% sp...scaled point(=1/65536pt)
% em
% ex
% in...inch (=72.26pt)
% sonstige: dd,cc,px
%
% mathmode: mu...math unit (=1/18em)
```

1.2.1 More geometries

```

\newgeometry{top=0.75in,bottom=1in,right=0.75in,left=0.75in}

\restoregeometry
```

1.2.2 Farben, Highlight

```
    \usepackage{tcolorbox}%required for \definecolor
    \usepackage{tcolorbox}%required for \definecolor

    \definecolor{purple2}{HTML}{660099}
    \definecolor{yellow}{HTML}{FCF434}
    \definecolor{purple}{HTML}{9C59d1}
    \definecolor{darkgray}{HTML}{2C2C2C}
    \definecolor{lightgray}{HTML}{C2C2C2}

    \newcommand{\graybf}[1]{\color{darkgray}\textbf{#1 }\color{black}}
```

1.2.3 Einzug

```
    \usepackage{changepage}%required for adjustwidth

    %%% In Document:
    \begin{adjustwidth}{2cm}{}
    ttext
    \end{adjustwidth}
```

1.2.4 Textlänge kürzen

\usepackage{truncate}

A really . . .

```
\truncate{64pt}{A really really long Text}
```

1.3 Datum

```
    \usepackage[ddmmyyyy]{datetime}
    \renewcommand{\dateseparator}{.}
```

1.4 Kopf- und Fußzeile

\usepackage{fancyhdr}

test Numerik Mitschrift 2023W V0.1-April 23, 2026

NameNameName Seite1 von

```
\pagestyle{fancy}%neededForChanging
\fancyhf{}%clearHeadersAndFooters
\fancyhead[L]{test}
\fancyhead[C]{Numerik Mitschrift 2023W}
\fancyhead[R]{V0.1-\today}
\fancyfoot[L]{NameNameName}
\fancyfoot[R]{Seite\thepage\hspace{3pt}von}
```

additional options for two-sided document

value	description
E	Even page
O	Odd page
L	Left field
C	Center field
R	Right field
H	Header
F	Footer

```
\fancypagestyle{two_sided}{
    \pagestyle{fancy}
    \fancyhf{}
    \fancyhead[RO]{RO}
    \fancyhead[LE]{LE}
    \fancyhead[LO,RE]{LO+RE}
    \fancyfoot[C]{C}
    \fancyfoot[LE,RO]{LE+RO}
}

\pagestyle{two_sided}
```

1.4.1 Seitenzahlanzeige

\usepackage{lastpage} provides \pageref{LastPage}.

roman page numbering

\pagenumbering{arabic}

1.5 Überschriften

\usepackage{titlesec}

titleformat, titlespacing beschreiben

```
    %required for \titleformat

    \titleformat{\chapter}[display]{\bfseries\Large} % format
        {\fontfamily{phv}\selectfont\scshape Kapitel \thechapter} % label
        {1ex} % sep
        {   \fontfamily{phv}\selectfont
            \bfseries\huge\centering
        } % before-code
        [\vspace{-0.2cm}] % after-code

    \titlespacing*{\chapter} {0pt}{*5}{*5}
    \titleclass{\chapter}{straight}
```

Kapitel 2

Images

\usepackage{graphicx}

Optional: set image path before by `\graphicspath{ {./img/} }`

TODO: make graphics

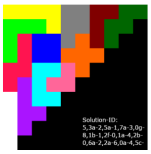

Figure 1: title

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

```
\lipsum[1]
\begin{figure}
        \centering
        \includegraphics[scale=0.3]{Ferres-diag}
        \caption{title}
\end{figure}
```

\usepackage{float}

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

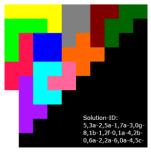

Figure 1: title

```
\lipsum[1]
\begin{figure}[H]
        \centering
        \includegraphics[scale=0.3]{Ferres-diag}
        \caption{title}
\end{figure}
```

Kapitel 3

Math

Online website for symbol name detection: https://detexify.kirelabs.org/classify.html

3.1 Special characters

```
\usepackage{mathtools}%required for \vcentcolon
```

```
        \newcommand{\defeq}{\vcentcolon=}% :=
        \newcommand{\eqdef}{=\vcentcolon}% =:
        \newcommand{\defaq}{\vcentcolon\Leftrightarrow}% :<=>
```

3.2 Math-Operator

```
\DeclareMathOperator{\matrixM}{M}
\DeclareMathOperator{\rk}{rank}
```

3.3 Equations

`\usepackage[leqno]{amsmath}` leqno ... left aligned equation numbering...

3.3.1 Multiline Equations

display long equations - first part will be aligned to the left and second part to the right.

$$\begin{multline*}
p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3\\
- 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3
\end{multline*}$$

```
\begin{multline*}
    p(x) = 3x^6 + 14x^5y + 590x^4y^2 + 19x^3y^3\\
    - 12x^2y^4 - 12xy^5 + 2y^6 - a^3b^3
\end{multline*}
```

3.3.2 Aligned Equations

Aligned Equation

$$\begin{aligned}
f(x) &= 42\\
g(x) &= 1-f(x)
\end{aligned}$$

```
\[
    \begin{aligned}
        f(x) &= 42\\
        g(x) &= 1-f(x)
    \end{aligned}
\]
```

Subequations with optional tag.

$$\begin{aligned}
&\forall x \: Q (x) \tag{$\bigstar$}\\
&\forall x \forall y \: (P(x,y))
\end{aligned}$$

```
\begin{subequations}
    \begin{align*}
        &\forall x \: Q (x) \tag{$\bigstar$}\\
        &\forall x \forall y \: (P(x,y))
    \end{align*}
\end{subequations}
```

3.4 Theorems

\usepackage{amsthm}

```
%% requires amsthrm package

\newtheorem{theorem}{Satz}[chapter]     %counter
\renewcommand{\thetheorem}{\arabic{chapter}.\arabic{theorem}}  %displaying
%%\newtheorem{name}[counter]{Printed output}
\newtheorem{defi}[theorem]{Definition}     %example
\newcommand{\defin}[2]{
    \ifSimplifiedVersion
        #1
    \else
        \color{purple}
        \begin{defi}
            \textbf{\fontfamily{txtt}\selectfont{#1}:} \normalfont \color{black}
            #2
        \end{defi}
        \color{black}
    \fi
}
```

3.5 Matrizen

3.6 Math alignment

\usepackage{mathtools}%automatically loads amsmath

smashoperator

$$Y = \bigcap_{X\subseteq S\subseteq G} S$$

$$Y = \bigcap_{X\subseteq S\subseteq G} S$$

```
\[
    Y = \smashoperator[l]{
        \bigcap_{X\subseteq S\subseteq G}
    } S
\]
\[
    Y = \smashoperator{
        \bigcap_{X\subseteq S\subseteq G}
    } S
\]
```

substack

$$\bigcap_{\substack{\mathcal{A}\subseteq \mathcal{P}(X)\\ A\subseteq \mathcal{A}}}$$

```
\[
    \bigcap_{\substack{
        \mathcal{A}\subseteq \mathcal{P}(X)\\
        A\subseteq \mathcal{A}\\
    }}
\]
```

move equation towards margin TODO: remove

$$\boxed{\begin{aligned}\lambda(f(A_n)) &\leq 2^{-n}\varepsilon\sum_{k=1}^{\infty}(b_k-a_k) \leq 2^{-n}\varepsilon\sum_{k=1}^{\infty}(b_k-a_k)+\sum_{j=2}^{\infty}a_j-b_{j-1}\\ &= 2^{-n}\varepsilon\left(b_1-a_1+\sum_{k=2}^{\infty}b_k-a_k+a_k-b_{k-1}\right) = 2^{-n}\varepsilon(b_1-a_1+\end{aligned}}$$

```
\begin{equation*}
    \begin{split}
        \hspace*{-2cm}\lambda(f(A_n))&\leq
            2^{-n}\varepsilon\sum_{k=1}^{\infty}{(b_k-a_k)}
            \leq
            2^{-n}\varepsilon\sum_{k=1}^{\infty}{(b_k-a_k)}
            +\sum_{j=2}^{\infty}{a_j-b_{j-1}}\\
        &=2^{-n}\varepsilon\Biggl(b_1-a_1 +
            \sum_{k=2}^{\infty}{b_k-a_k+a_k-b_{k-1}}\Biggr)=
            2^{-n}\varepsilon(2n)=2^{1-n}n\varepsilon\\
    \end{split}
\end{equation*}
```

Kapitel 4

Auflistungen

4.1 enumitem

\usepackage{enumitem} is required for the remainder of the section.

4.1.1 Einzug in Liste

(Regel 1)

(R 2)

```
\begin{itemize}[leftmargin=4cm]
    \item[(Regel 1)]
    \item[(R 2)]
\end{itemize}
```

4.1.2 Andere Symbole, Abstände, Horizontale Bulletpoints

Andere Symbole: `label=(\roman*)` für kleine römische Ziffern in Runden Klammern. `\Roman*` für römische Ziffern.

Small roman numbers

(i) one

(ii) two

```
Small roman numbers
\begin{enumerate}[label=(\roman*)]
    \item one
    \item two
\end{enumerate}
```

Horizontale Liste: Wie Fließtext, `leftmargin` wie oben funktioniert hier nicht. "\\" für line-breaks. inline

`\usepackage[inline]{enumitem}`

Axioms (T1) $\emptyset, X \in \tau$ (T2) second (T3) third

```
\usepackage[inline]{enumitem}%for enumerate*
    (i.e. horizontal lists)
Axioms
\begin{enumerate*}
        \item[(T1)] $\emptyset, X\in \tau$
        \item[(T2)] second
        \item[(T3)] third
\end{enumerate*}
```

Abstände:

Roman Numbers

(I) one

(II) two

```
Roman Numbers
\begin{enumerate}[label=(\Roman*),
    itemsep=0pt, topsep=0pt]
    \item one
    \item two
\end{enumerate}
```

4.1.3 Liste mit anderer Zahl anfangen

(iv) $K(() = -1$ and $K()) = 1$

(v) $K(\forall) = 1$, $K(\neg) = 0$ and $K(\to) = -1$

```
    \begin{enumerate}[label=(\roman*)]
        \setcounter{enumi}{3}
        \item $K(() = -1$ and $K()) = 1$
        \item $K(\forall) = 1$, $K(\lnot) = 0$
            and $K(\to) = -1$
    \end{enumerate}
```

For geschachtelete Listen: `enumii`, `enumiii` oder `enumiv` verwenden.

4.2 Tabelle

Tabelle mit vertikaler Linie durch `\hline`

value	state
0	off
1	on
2	blinking

```
\begin{center}
    \begin{tabular}{c|l}
        value & state\\
        \hline
        0 & off\\
        1 & on\\
        2 & blinking
    \end{tabular}
\end{center}
```

Kapitel 5

Bibliography

```latex
\usepackage[backend=biber,style=alphabetic]{biblatex}
\addbibresource{BA-ref.bib}
```

```latex
\cite{id}
\hline
\printbibliography
```

Kapitel 6

Sourcecode

```
\usepackage{listings}
\usepackage{color}

\definecolor{mygreen}{rgb}{0,0.6,0}
\definecolor{mygray}{rgb}{0.5,0.5,0.5}
\definecolor{mymauve}{rgb}{0.58,0,0.82}

\lstset{
backgroundcolor=\color{white}, % choose the background color; you must add \usepackage{color} or \usepackage{xcolor}; should come as last argument
basicstyle=\footnotesize,      % the size of the fonts that are used for the code
breakatwhitespace=false,       % sets if automatic breaks should only happen at whitespace
breaklines=true,               % sets automatic line breaking
captionpos=b,                  % sets the caption-position to bottom
commentstyle=\color{mygreen}, % comment style
deletekeywords={...},          % if you want to delete keywords from the given language
escapeinside={\%*}{*)},        % if you want to add LaTeX within your code
extendedchars=true,            % lets you use non-ASCII characters; for 8-bits encodings only, does not work with UTF-8
firstnumber=1000,              % start line enumeration with line 1000
frame=single,                  % adds a frame around the code
keepspaces=true,               % keeps spaces in text, useful for keeping indentation of code (possibly needs columns=flexible)
keywordstyle=\color{blue},     % keyword style
language=Octave,               % the language of the code
morekeywords={*,...},          % if you want to add more keywords to the set
numbers=left,                  % where to put the line-numbers; possible values are (none, left, right)
numbersep=5pt,                 % how far the line-numbers are from the code
numberstyle=\tiny\color{mygray}, % the style that is used for the line-numbers
rulecolor=\color{black},       % if not set, the frame-color may be changed on line-breaks within not-black text (e.g. comments (green here))
showspaces=false,              % show spaces everywhere adding particular underscores; it overrides 'showstringspaces'
showstringspaces=false,        % underline spaces within strings only
showtabs=false,                % show tabs within strings adding particular underscores
stepnumber=2,                  % the step between two line-numbers. If it's 1, each line will be numbered
stringstyle=\color{mymauve}, % string literal style
tabsize=2,                   % sets default tabsize to 2 spaces
title=\lstname                 % show the filename of files included with \lstinputlisting; also try caption instead of title
}
```

Kapitel 7

Fun Extras

7.1 If-Then Unterscheidungen

variable or ifthenelse

```
\usepackage{ifthen}%for "ifthenelse"
\newif\ifSimplifiedVersion
%\SimplifiedVersiontrue %uncomment this line for simplified version
```

```latex

    \ifSimplifiedVersion
        %simplified
    \else
        %not simplified
    \fi

    %ifthenelse
    \ifthenelse{\equal{#3}{}}{wahr}{
        falsch
    }
```

Falls #3 allerdings beispielsweise auflistungen (itemize, enumerate) enthält, entsteht eine fehlermeldung. Daher ein possible workaround:

```latex
    %TODO check for packages

    \newcounter{propcount}[chapter]
    \renewcommand{\thepropcount}{Theorem \theHchapter.\arabic{propcount}}

    \newtheorem{propo}[propcount]{}
    \newcommand{\prop}[3]{
        %Input: 1-Name, 2-statement, 3-proof
        \ifSimplifiedVersion
            Prop: #1 - #2

        \else
            \color{purple}
            \begin{propo}
                \textbf{\fontfamily{txtt}\selectfont{#1}:} \normalfont\color{black}
                #2
                \ifx\foo#3\foo
                \else
                    \begin{proof}
                        #3\vspace{5pt}
                    \end{proof}
                \fi
            \end{propo}
            \color{black}
        \fi
    }
```

7.2 Inhaltsverzeichnis / Table of contents

7.2.1 Veringern des vertikalen Abstandes im / vorm Inhaltsverzeichnis

```latex
    \usepackage{tocloft}

    \setlength{\cftbeforetoctitleskip}{0em} % vor titel und inhaltsverzeichnis
    \setlength{\cftaftertoctitleskip}{0em} % zwischen titel und inhaltsverzeichnis
```

7.2.2 Eigene Verzeichnisse

```latex
    \usepackage{tocloft}
    \newcommand{\listNAMEname}{\Large List of NAME}
    \newlistof{NAME}{bezeichnung}{\listNAMEname}


    \begin{document}
        \section{hallo}
        \addcontentsline{bezeichnung}{section}{\thesection }
```

```
        \listofNAME
    \end{document}
```

List of NAME

0.1 14